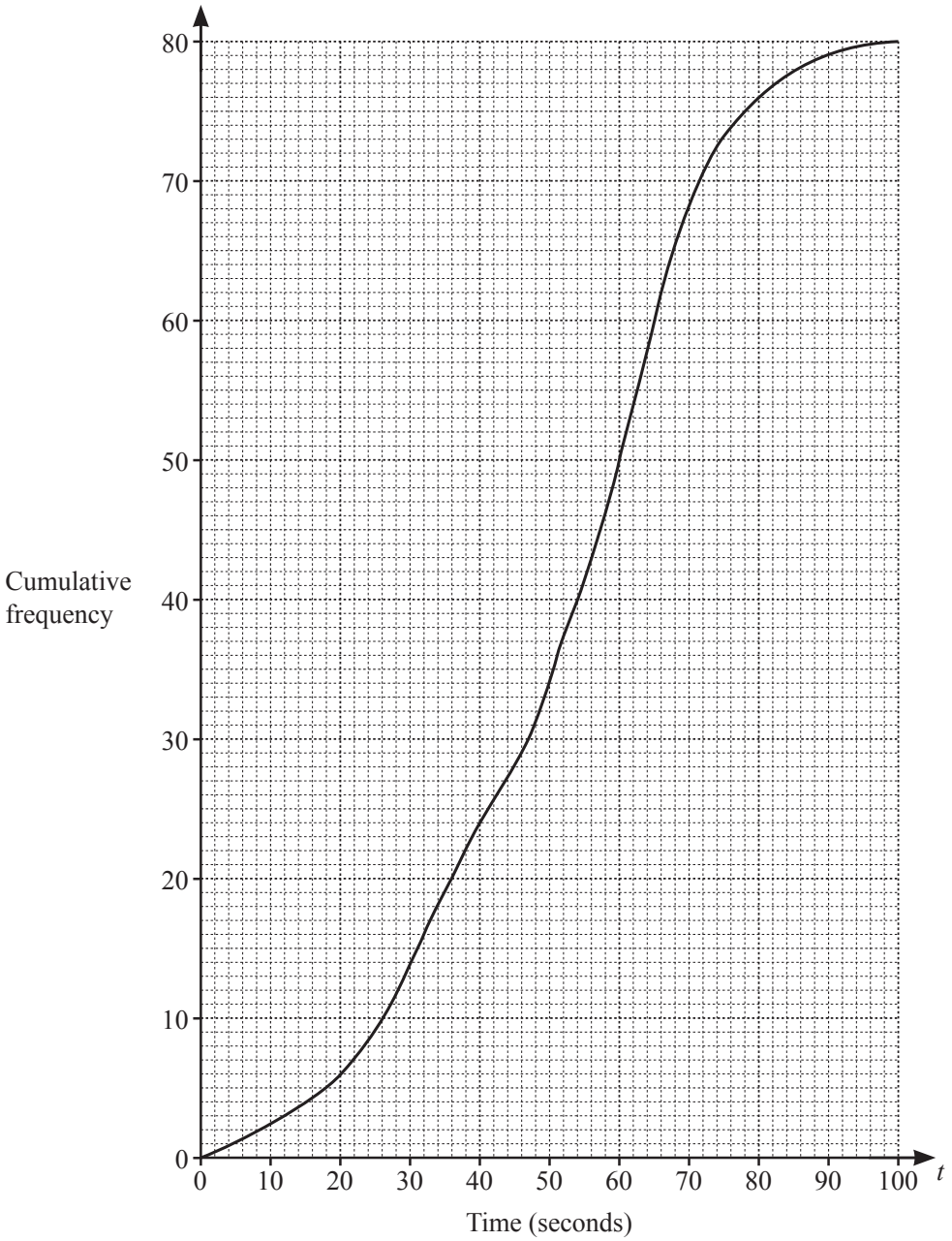
2 The cumulative frequency diagram shows information about the time taken, t seconds, for a group of girls to each solve a maths problem.



(a) Use the cumulative frequency diagram to find an estimate for

(i) the median,

..... s [1]

(ii) the interquartile range,

..... s [2]

(iii) the 20th percentile,

..... s [1]

(iv) the number of girls who took more than 66 seconds to solve the problem.

..... [2]

(b) **(i)** Use the cumulative frequency diagram to complete the frequency table.

Time (t seconds)	$0 < t \leqslant 20$	$20 < t \leqslant 40$	$40 < t \leqslant 60$	$60 < t \leqslant 80$	$80 < t \leqslant 100$
Frequency	6				4

[2]

(ii) Calculate an estimate of the mean time.

..... s [4]

(c) A group of boys solved the same problem.
The boys had a median time of 60 seconds, a lower quartile of 46 seconds and an upper quartile of 66 seconds.

(i) Write down the percentage of boys with a time of 66 seconds or less.

..... % [1]

(ii) Howard says

The boys' times vary more than the girls' times.

Explain why Howard is incorrect.

.....

..... [2]